

TIØ4145 Corporate Finance

Exercise 6: Capital structure – Part 1

Deadline: Monday October 6th, 2025, 23:59 – Grading: Passed/Failed

Your answers should be submitted as a PDF file to BlackBoard before the deadline, either written in MS Word, LaTeX or handwritten with a high-quality scan. You are also welcome to upload additional Excel computations if you find it necessary. Guidance to this exercise is given on Thursday October 2nd, 8:15 – 10:00 in A32. Solutions to the problems will be posted on BlackBoard after the deadline.

Relevant Curriculum

- Lecture notes from lecture given on Monday September 29th.
- Berk and DeMarzo, Corporate Finance 5th edition: Chapters 14 and 15.

Tasks

Chapter 14 – Capital Structure in a Perfect Market

1. Yerba Industries is an all-equity firm whose stock has a beta of 0.90 and an expected return of 11.11%. Suppose it issues new risk-free debt with a 6% yield and repurchases 55% of its stock. Assume perfect capital markets.

- a. What is the beta of Yerba stock after this transaction?
- b. What is the expected return of Yerba stock after this transaction?

Suppose that prior to this transaction, Yerba expected earnings per share this coming year of \$5, with a forward P/E ratio (that is, the share price divided by the expected earnings for the coming year) of 9.

- c. What is Yerba's expected earnings per share after this transaction? Does this change benefit shareholders? Explain.

- d. What is Yerba's forward P/E ratio after this transaction? Is this change in the P/E ratio reasonable? Explain.
2. Hardmon Enterprises is currently an all-equity firm with an expected return of 12%. It is considering a leveraged recapitalization in which it would borrow and repurchase existing shares.
 - a. Suppose Hardmon borrows to the point that its debt-equity ratio is 0.50. With this amount of debt, the debt cost of capital is 6%. What will the expected return of equity be after this transaction?
 - b. Suppose instead Hardmon borrows to the point that its debt-equity ratio is 1.50. With this amount of debt, Hardmon's debt will be much riskier. As a result, the debt of capital will be 8%. What will the expected return of equity be in this case?
 - c. A senior manager argues that it is in the best interest of the shareholders to choose the capital structure that leads to the highest expected return for the stock. How would you respond to this argument?
 3. You are CEO of a high-growth technology firm. You plan to raise \$170 million to fund an expansion by issuing either new shares or new debt. With the expansion, you expect earnings next year of \$27 million. The firm currently has 10 million shares outstanding, with a price of \$74 per share. Assume perfect capital markets.
 - a. If you raise the \$170 million by selling new shares, what will the forecast for next year's earnings per share be?
 - b. If you raise the \$170 million by issuing new debt with an interest rate of 10%, what will the forecast for next year's earnings per share be?
 - c. What is the firm's forward P/E ratio (that is, the share price divided by the expected earnings for the coming year) if it issues equity? What is the firm's forward P/E ratio if it issues debt? How can you explain the difference?

Exam Question – 2024

EcoNor Ventures, an ambitious company leading the green transition, has assets with a market value of \$600 million, \$70 million of which are cash. The company has debt totaling \$250 million and currently has 7 million shares outstanding. Assume perfect capital markets.

- a. What is EcoNor's current stock price?
- b. If EcoNor distributes the cash of \$70 million as a share repurchase, what will its stock price be after the share repurchase?
- c. How many shares will remain outstanding after the repurchase?

Chapter 15 – Debt and Taxes

1. Suppose the corporate tax rate is 40%. Consider a firm that earns \$2500 before interest and taxes each year with no risk. The firm's capital expenditures equal its depreciation expenses each year, and it will have no changes to its net working capital. The risk-free interest rate is 4%.
 - a. Suppose the firm has no debt and pays out its net income as a dividend each year. What is the value of the firm's equity?
 - b. Suppose instead the firm makes interest payments of \$1600 per year. What is the value of equity? What is the value of debt?
 - c. What is the difference between the total value of the firm with leverage and without leverage?
 - d. The difference in part (c) is equal to what percentage of the value of the debt?
2.  PMF, Inc., is equally likely to have EBIT this coming year of \$7 million, \$16 million, and \$25 million. Its corporate tax rate is 38%, and investors pay a 25% tax rate on income from equity and a 40% tax rate on interest income.
 - a. What is the effective tax advantage of debt if PMF has interest expenses of \$6 million this coming year?
 - b. What is the effective tax advantage of debt for interest expenses in excess of \$25 million? (Ignore carryforwards.)
 - c. What is the expected effective tax advantage of debt for interest expenses that are above \$7 million but below \$16 million? (Ignore carryforwards.)
 - d. What level of interest expense provides PMF with the greatest tax benefit?

Hint for (d): Using Excel, you can calculate the average FCF for the three scenarios with a series of interest payments with 1 million increment, using the following logic. Why?

IF Pretax – Interest > 0

$$(\text{Pretax} - \text{Interest}) * (1 - \tau_C) * (1 - \tau_E) + \text{Interest} * (1 - \tau_I)$$

ELSE

$$(\text{Pretax} - \text{Interest}) * (1 - 0) * (1 - \tau_E) + \text{Interest} * (1 - \tau_I)$$

Exam Question – 2024

Consider Xpertise AI, a company claiming to have mastered everything from social media to space exploration algorithms. It is expected to generate a free cash flow of \$8 million in the coming year, with this cash flow projected to grow at 3% annually. Xpertise AI has an equity cost of capital of 13%, a debt cost of capital of 7%, and faces a 21% corporate tax rate. The company also maintains a constant debt-to-equity ratio of 0.5.

- i. Calculate Xpertise AI's Weighted Average Cost of Capital (WACC).
- ii. Determine the value of Xpertise AI as an unlevered company, ignoring the effects of debt.
- iii. Calculate the value of Xpertise AI's interest tax shield.

Exam Question – 2017 (Voluntary)

As a financial consultant, you are given the following information about two companies, one levered and one unlevered.

Copy the table below and fill in the missing figures. Show the steps you use to calculate them. Assume that the corporate tax rate is 40% and all cash flows are perpetual.

	Unlevered	Levered
Before tax operating income	\$10 000	\$10 000
Interest on debt	0	\$2 500
Cost of capital (WACC)		16%
Cost of Equity capital		
Cost of Debt capital		
Value of equity		\$22 500
Value of debt		
Value of firm		

References

Berk, J. and DeMarzo, P. (2017). Corporate Finance, Fifth Edition, Pearson Education, Essex, England.